

Service Operations Analytics

From incident audit trails to reliable reporting and demand planning.

4.66%

NYC test WAPE

97.6%

coverage of nominal 90% band

3 interactive views

incident, demand, resolution

Problem and business decision

A backlog total alone cannot explain workload changes, missing ownership, or future demand. The project supports queue investigation and capacity scenarios while exposing data quality and forecast uncertainty.

Data and engineering

A fixed 2016 UCI incident audit log is replayed with simulated duplicates, corrections and late deliveries. Current NYC 311 data supplies daily aggregate demand. Checks gate publication; immutable batches, versioned Parquet facts and dimensions, Spark SQL and Athena support reconciliation. S3 and CloudFront serve public aggregates.

Models and operating evidence

Time-ordered selection, calibration and a 42-day test compare weekday baselines with seasonal regression. Data cutoff: 2026-09-19; results above are a dated snapshot. A weather/calendar challenger remains separate. Published forecasts are retained for later scoring; retrospective charts are labeled separately. A censor-aware resolution model improves test probability error, but does not establish better resolution-time error.

What a visitor can use

Filter boroughs, compare uncertainty and error by weekday/lead, and move a capacity slider to see the complete demand-gap trade-off. Inspect backlog inflows/outflows and critical-versus-moderate survival curves. Download aggregate data and evaluations; no prediction endpoint or account is required.

Cost and limits

Planning allowance: \$2-5/month at modest traffic, not a measured monthly bill or a hard cap. Scheduled serverless work avoids always-on compute. No measured staffing savings or adoption claim. This is daily batch forecasting, not streaming; source revisions, small subgroups, and dependent errors limit conclusions. The 44 MB incident workload does not demonstrate distributed scale.

Live: d17q1whdf7htno.cloudfront.net/forecast/

Code and definitions: github.com/RyanSingh0/service-operations-analytics-portfolio

Sources: UCI DOI 10.24432/C57S4H (CC BY 4.0); NYC Open Data 311 (erm2-nwe9).